



Less Water, Better Living

Lessons from the 50L Home Pilot on Scaling
Household Water Efficiency



PART ONE

Why 50L?



Premise

Reduce water use by making sustainability irresistible

We're not *asking* households to use less water.

We're supporting performance-driven design and behavior change strategies that make home routines simpler and more convenient – water savings naturally follow.



Who we are

50L Home: a public-private platform for innovation

Co-Chairs



Members



Partners



Secretariat



Collaboration

Private sector: design and innovation

Utilities: service delivery

Government: regulation

Households: use behaviors

OUR MISSION

Make 50-liter living the standard through innovation that improves quality of life.



Problem

Barriers to affordable and sustainable living



Water scarcity

Snowpack in the Colorado River watershed is the smallest on record – a fifth of its average levels.

Significant room to save dwindling water resources through indoor efficiency



Affordability

18% of Californians pay more than 1.5% the median household income for their monthly water bills (*source*: PPIC)

Every gallon saved cuts utility bills



Energy-water nexus

Hot water accounts for 18% of home energy use – second largest driver of energy bills (*source*: Department of Energy)

Reduced hot water consumption lowers energy bills.

What we've learned

Two years, 31 homes, fixture-level data

LA Pilot

Two years, three phases, fixture-level data

31

households

15

retrofit / 16 control

2 yrs

of measurement

6

submeters per home

M E T H O D O L O G Y

- Whole-home Flume monitors plus Droople fixture submeters at six points per home
- Minute-by-minute flow rates and water temperatures
- Capture behavioral data on whole-home water use

PHASE 1 4 months

Baseline

Existing fixtures and appliances measured across all 32 homes.

PHASE 2 12 months

Market-available retrofits

Latest products you can buy today Kohler/IKEA fixtures, Electrolux W/D, P&G consumables.

PHASE 3 12 months

Prototype products

Next-generation fixtures (~30% below code), prototype appliances, recirculating shower test.

LA Pilot

High-frequency monitoring at the appliance level



Headline Results



21

**gallons / person /
day**



56%

**less water use than LA
household average**



4.4

/ 5 satisfaction

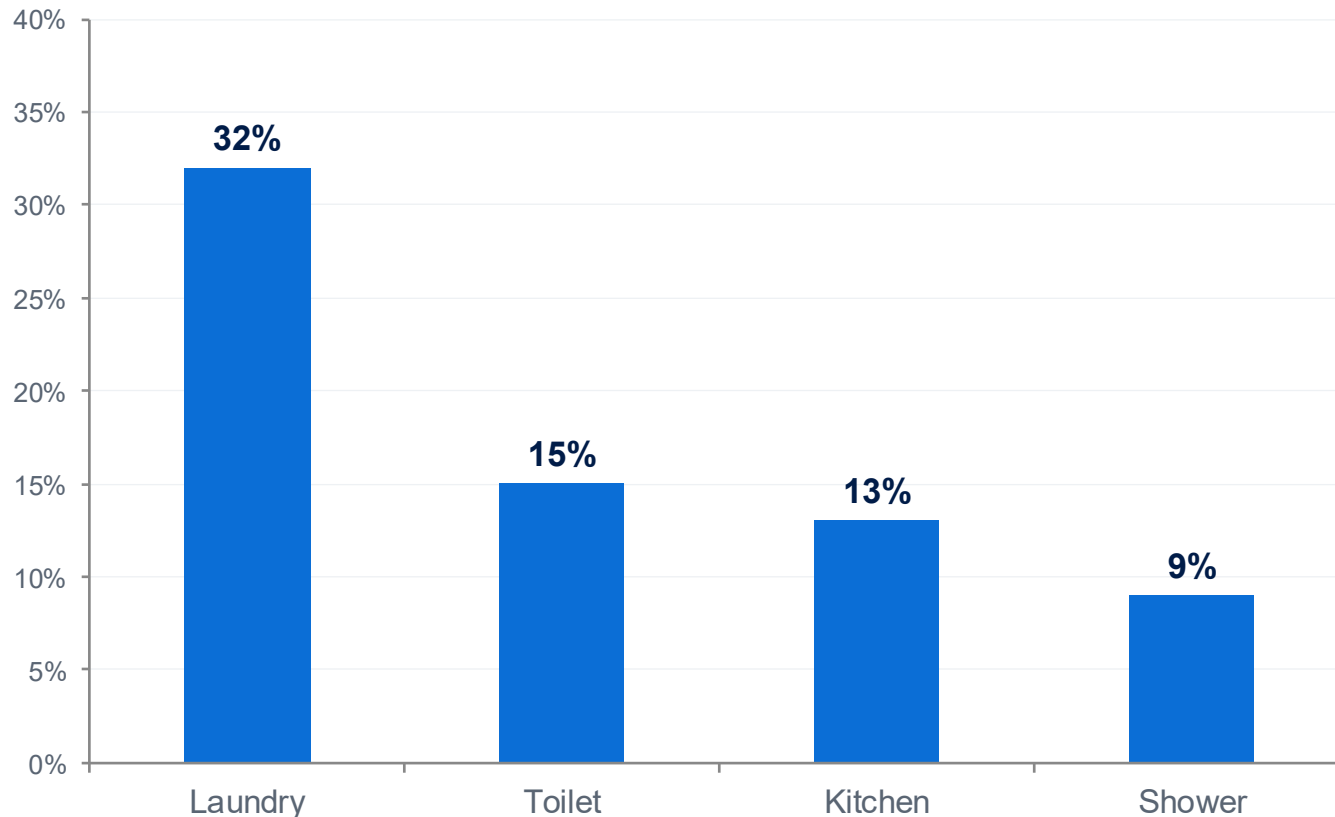


18%

**reduction in energy
use relative to
baseline**

By the Numbers

Reductions in water use by category

**CUSTOMER BENEFITS****Kitchen (HE Faucet/Dishwasher + Soaps)**

- ✓ No pre-rinsing
- ✓ Minimal soaking/scrubbing

Laundry (HEFL Washer + Coldwater Detergent)

- ✓ Fewer loads
- ✓ No sorting
- ✓ No hot water required – 18% energy savings

Shower (HP Showerhead + 2-in-1 Products)

- ✓ Kohler 1.25 GPM showerheads reduced streamflow with no behavioral compensation

Toilet (0.8 GPF Toilet + HP Toilet Paper)

- ✓ Reduced double flushing
- ✓ Fewer clogs

Insight 1

Pairing hardware and consumables enables behavior change

No single product delivered the full reduction. The savings came from pairings: efficient hardware plus high-performing consumables.



HE Dishwasher + Cascade + Dawn
Powerwash

44%

kitchen sink water reduction

Pre-rinsing essentially eliminated. The appliance enables the behavior change; the consumable removes the friction.



HE Washer + Tide Coldwater Clean

54%

laundry hot water reduction

Cold-water washing became default, not the exception. Sorting by color became unnecessary. Compounded water and energy savings.

Insight 2

Choice architecture matters



Dual options increase water use.

Dual-flush toilets actually increased water use by 35%.

Participants reported confidence in using them correctly but defaulted to the higher-volume button.

Rinse booster features increased water use by 24%



Participants stick to default options.

Even highly engaged participants left savings on the table:

18 of 26 never tried eco dishwasher cycles

24 of 26 never tried the dryer refresh feature

Additional savings achievable through education and prompt

Business Case

Positive ROI and durable savings

\$5–7K

retail retrofit cost

Whole-home: dishwasher, kitchen faucet, washer, dryer, toilet, showerhead, bathroom faucets, plus a year of consumables.

34,500

gallons saved/year

Per household, every year — roughly equivalent to removing 1,000 sq ft of irrigated turf.

\$890

\$/year

Cost savings per household, every year

BENEFITS RELATIVE TO OUTDOOR EFFICIENCY

**Stable reductions in demand**

Indoor savings don't fluctuate with rainfall or season.

**Equitable**

Reaches rentals and multi-family housing, customer segments outdoor programs miss entirely.

**Direct-install ready**

Fits inside existing utility rebate and CAP infrastructure.

Key takeaways

Key takeaways by stakeholder and future work

**WATER/ POWER UTILITIES****Indoor water efficiency delivers reliable water & energy savings.**

- How can utilities integrate top-line efficiency products into existing rebate and incentive structures?
- What incentive models optimize the ratio of program expenses to household uptake?

**HOUSEHOLDS****Making 50L feel “irresistible”**

- What use behaviors drive actual water savings and how can design choices influence them?
- What additional savings can be achieved through nudges and education?

**GOVERNMENTS****Making 50L the standard for housing**

- How do we work with developers to build at 50L specs?
- What reforms and code/permit pathways are required to enable broader adoption of 50L Home technology?

**INDUSTRY****Performance drives market success.**

- How can we continue to innovate to close the final stretch from 79L to 50L/day?

Contact Us



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Learn more: <https://50lhome.org>

Annexes

LA Pilot

Cost-benefit analysis

Product	Payback	Water Savings	10-Year Dollar Savings
Laundry (Washer + Detergent)	~3 years	56,756 liters 14,990 gallons	\$1,516
Kitchen (Dishwasher + Products)	~2 years	66,446 liters 17,560 gallons	\$1,357
Combined Indoor Bundle (Kitchen + Laundry)	As early as 2 years	123,000 liters 32,490 gallons	\$2,873

LA Pilot

Appliances used in LA pilot

Phase 2		Phase 3	
Kitchen  Low-flow kitchen faucet 1.5 gpm  High efficiency dishwasher 3.3 gal/cycle 		Kitchen  Ultra efficient kitchen faucet 1.0 gpm  Dishwasher with Smart Boost and Lux Care 	
Bathroom  Low-flow showerhead 1.5 / 1.8 gpm  Low-flow bathroom faucet 1.0 / 1.2 gpm  High efficiency (1.0 gpf) or dual-flush toilet (1.1 / 1.6 gpf) 		Bathroom  Dual flush toilet 1.0/0.8 gpf  Ultra efficient showerhead 1.25 gpm  Shower booster (on/off button)  Faucet mister  Recirculating shower and autodiverter 	
Laundry Room  High Efficiency Washer and Dryer 		Laundry Room  Perfect Steam Washer and Dryer with LuxCare Plus Wash and SmartBoost 	